

Virtual Prejudice in Childhood: Impulsive Avoidance Toward Racial Out-Group Avatars

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ABSTRACT

Adults maintain larger interpersonal distances from social out-groups due to implicit biases, yet it remains unclear if this emerges in childhood. We utilized Virtual Reality (VR) to investigate personal space regulation in 30 Japanese children (aged 8–16). Participants performed a Stop-Distance Task with avatars varying in race (Asian vs. White), sex (Female vs. Male), and approach direction (Active vs. Passive). Results revealed a significant main effect of race: children maintained larger distances from White (out-group) avatars than Asian (in-group) avatars, suggesting that the implicit biases manifest early in development. Additionally, we found a significant interaction between age and approach direction; while the distance in Passive condition remained stable across all ages, in Active condition, older children approached avatars significantly closer than younger children. These findings suggest that interpersonal distance and approach dynamics may be important considerations when designing VR interactions for children, particularly in contexts involving unfamiliar social agents.

Index terms: Empirical studies in HCI, implicit biases, personal space, racial familiarity

1 INTRODUCTION

Personal space serves as a non-verbal buffer that regulates social interaction, expanding or contracting based on comfort and familiarity [1]. In adults, this boundary is affected by social group membership. Research using virtual reality (VR) indicates that implicit biases can manifest as an impulsive avoidance reaction, leading individuals to automatically maintain larger distances from racial out-group members [2]. Theoretical frameworks such as the Reflective-Impulsive Model (RIM) suggest this phenomenon is driven by automatic processes rather than reflective deliberation [3].

However, the developmental trajectory of these spatial biases remains under-explored. While racial categorization is established by early childhood [4], [5], it is unclear whether the specific implicit avoidance behaviors observed in adults manifest similarly in children and adolescents, or how these spatial biases evolve as social agency develops. To address this gap, this study investigated the effects of racial familiarity and sex on personal space regulation in Japanese children. By employing a within-subjects design, we aim to determine if children demonstrate an implicit in-group preference through increased interpersonal distance when interacting with racial out-group avatars.

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2 METHOD

2.1 Participants

A total of 35 Japanese children (19 females, 16 males) with a mean age of 12.91 years ($SD = 2.20$, range = 8–16) were recruited and participated in the experiment. Five participants were excluded from the analysis due to calibration failure ($n=2$) or system errors ($n=3$). The final sample comprised 30 participants (16 females, 14 males) with a mean age of 13.03 years ($SD = 2.21$, range = 8–16).

2.2 Materials

The VR environment was presented via a head-mounted display (HMD; VIVE Pro Eye, HTC, Taiwan). The visual stimuli were rendered on a workstation (Alienware x16 R2, Dell, Round Rock, TX) equipped with a NVIDIA GeForce RTX 4080 GPU, 32GB of RAM, and an Intel Core Ultra 9 185H CPU, running on the Windows 11 operating system. We developed the VR environment and avatar models using Unity (v.2021.3.4f) and Daz3D (v.4.21). A series of humanoid virtual avatars (Asian female: Kei HD for Genesis 9, White female: HID Ashley for Genesis 9, Asian male: E3D Aroon HD for Genesis 9, White male: Malcolm HD for Genesis 8.1 male) was used, featuring a neutral appearance and facial characteristics. The avatar's body was animated using a standard walking motion (Walk Animset Basic) and dressed in a gray tracksuit (dForce Tracksuit Outfit for Genesis 9, 8, and 8.1) to maintain a neutral and realistic representation. To ensure consistent orientation, the avatar was programmed to continuously face the participant in real time. Participants' responses were collected via a USB response button (Ne Fashion), held in their dominant hand. Gaze data were collected using the eye-tracking system built into the HMD, using SRanipal software (v.1.3.6.8).

2.3 Experimental Design

The study employed a $2 \times 2 \times 2$ within-subjects factorial design to examine the effects of familiarity. The within-subjects factors included race condition (Asian vs. White), Sex (Female vs. Male), and approach direction (passive vs. active). Continuous covariates of participants age were included in the analysis to examine individual differences.

Participants completed a Stop-Distance task, a widely used and reliable technique for assessing personal [1]. Personal space was computed as the horizontal (floor-parallel) distance between the participant's head-tracked position (as measured by the HMD) and the waist position of the virtual avatar. We defined personal space as the distance between the participant and the avatar when the participant either no longer wished the avatar to approach or no longer felt comfortable approaching the avatar.

The task included two conditions: passive and active approach (see Fig. 1). In each trial, participants began at the center of a red cross on the virtual floor, with the avatar positioned 5 meters away. Participants returned to the center of the cross after each trial. In the

passive approach condition, the avatar moved toward the participant at a constant speed of 95.87 cm/s. Participants were instructed to press a button held in their dominant hand as soon as they felt the avatar was too close for comfort. If the participant did not feel uncomfortable regardless of the avatar's proximity, they were instructed to press the button once the avatar's position fully overlapped with their own, in which case the interpersonal distance was recorded as 0 meters. In the active approach condition, the avatar remained still while the participant walked toward the avatar at their own pace, pressing the button to signal when they felt too close. After each trial, participants were instructed to return to the center of the virtual cross before proceeding. As in the passive approach condition, if participants felt no discomfort at any distance, they were instructed to stop and press the button once their own position fully overlapped with the avatar, and the interpersonal distance was recorded as 0 meters.



Figure 1: Active approach condition (left), and passive approach condition (right).

2.4 Experiment Procedure

All participants provided written informed consent. Participants were then fitted with a VR headset and provided a USB button in their dominant hand. An eye-tracking calibration was performed to ensure accurate data collection of eye movements.

2.4.1 Practice Trial

Following the calibration, participants completed a practice trial to familiarize themselves with the VR environment and the Stop-Distance Task procedures. This practice trial included a single instance of both the passive approach and active approach conditions of the Stop-Distance Task, as well as a brief interaction phase. To avoid any potential confounding effects of habituation to a specific avatar, a different virtual avatar model was used for the practice trial.

2.4.2 Main Trial

After calibration, participants performed the Stop-Distance Task. The task was comprised of 24 trials in total, with 12 trials each for the passive approach and active approach conditions. The order of these approach conditions and the avatars were counterbalanced across participants.

2.5 Statistical Analysis

Linear mixed-effects models were conducted using restricted maximum likelihood (REML) estimation in R (version 4.3.0) with the lme4 [6] and lmerTest [7] packages. A linear mixed-effects model was fitted to the interpersonal distance data with Approach Direction (Active vs. Passive), Avatar Race (Asian vs. White), Avatar Sex (Female vs. Male), and Age (scaled) as fixed effects, and Participant ID as a random intercept including a random slope for Approach Direction.

3 RESULTS

We found a significant main effect of Avatar Race ($t(646) = -3.48, p < .001$). Participants maintained significantly larger interpersonal distances when interacting with White (out-group) avatars ($M =$

1.18 m, $SE = 0.12$) compared to Asian (in-group) avatars ($M = 1.10$ m, $SE = 0.12$). There was also a significant main effect of Avatar Sex ($t(646) = -3.08, p = .002$). Participants maintained larger distances from Male avatars ($M = 1.17$ m, $SE = 0.12$) compared to Female avatars ($M = 1.11$ m, $SE = 0.12$) (see Fig. 2).

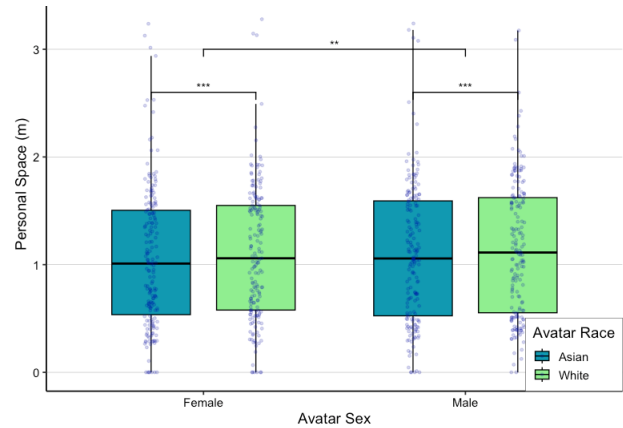


Figure 2: Effects of avatar race and sex on personal space (***: $p < .001$, **: $p < .01$).

A significant main effect of Approach Direction was observed ($t(75) = -2.91, p = .005$). Overall, participants approached the avatar more closely in the Active condition ($M = 1.06$ m) compared to the Passive condition ($M = 1.23$ m). Furthermore, a significant Age $\times$ Approach Direction interaction was found ($\beta = -0.12, p = .041$). While personal space in the Passive condition remained relatively stable across the age range (slope $\beta \approx -.03, p = .80$), the distance in the Active condition showed a steeper negative trend (slope $\beta \approx -.11, p = .36$), though the simple slopes themselves did not reach significance due to sample size.

4 CONCLUSION

This study investigated spatial biases in children using a VR environment. Our results confirm that implicit racial biases manifest early in development, as Japanese children maintained significantly larger interpersonal distances from out-group (White) avatars compared to in-group (Asian) ones. Additionally, we observed a distinct developmental trajectory regarding social agency: while the passive reaction to being approached remained stable across ages, the willingness to active approach an avatar increased significantly as children grew older. These findings demonstrate that personal space in children is a dynamic boundary shaped by both social group membership and developmental maturity. Consequently, developers of VR applications for education and healthcare must account for these baseline spatial biases to ensure physically and psychologically safe embodied experiences for young users.

ACKNOWLEDGMENTS

The authors wish to thank members of the Terada Laboratory, specifically Y. Noma, K. Noda, T. Hasegawa, and S. Fujii, for contributing to the development of the virtual reality system, as well as A. Fujii for assistance with data collection.

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